to be a lubrication failure. Good to know, but what to do about it? Plainly, it would be impossible to overtake *Voyager* with an oilcan.

The engineers wondered whether they could restart the failed actuator by alternate heating and cooling; maybe the resulting thermal stresses would induce the components of the actuator to expand and contract at different rates and unjam the system. They tested this notion with specially manufactured actuators in the laboratory, and then jubilantly found that in this way they could start the scan platform up again in space. Project personnel also devised ways to diagnose any additional trend toward actuator failure early enough to work around the problem. Thereafter, *Voyager 2*'s scan platform worked perfectly. All the pictures taken in the Uranus and Neptune systems owe their existence to this work. The engineers had saved the day again.

*Voyagers 1* and 2 were designed to explore the Jupiter and Saturn systems only. It is true that their trajectories would carry them on past Uranus and Neptune, but officially these planets were never contemplated as targets for *Voyager* exploration: The spacecraft were not supposed to last that long. Because of our wish to fly close to the mystery world Titan, *Voyager 1* was flung by Saturn on a path that could never encounter any other known world; it is *Voyager 2* that flew on to Uranus and Neptune with brilliant success. At these immense distances, sunlight is getting progressively dimmer, and the radio signals transmitted to Earth are getting progressively fainter. These were predictable but still very serious problems that the JPL engineers and scientists also had to solve.

Because of the low light levels at Uranus and Neptune, the *Voyager* television cameras were obliged to take long time exposures. But the spacecraft was hurtling so fast through, say, the Uranus system (at about 35,000 miles per hour) that the image would have been smeared or blurred. To compensate, the entire spacecraft had to be moved during the time exposures to cancel out the motion, like panning in the direction opposite yours while taking a photograph of a street scene from a moving car. This may sound easy, but it's not: You have to neutralize the most innocent of